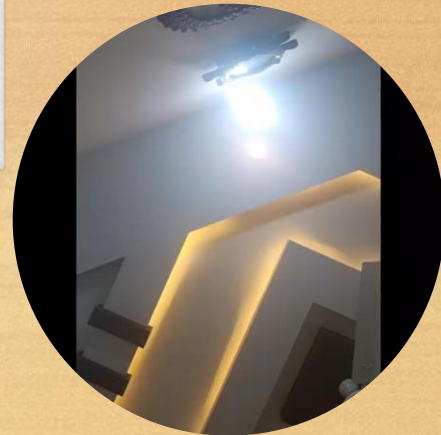


Python

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Introduction to tuple

- A Tuple is a data type in Python.
- It is used to store multiple values in a single variable.
- Tuples are ordered and immutable.
- Immutable means values cannot be changed after creation.

Example:

```
my_tuple = (10, 20, 30, 40)
```



Characteristics of Tuple

- Ordered collection
- Immutable (cannot be changed)
- Allows duplicate values
- Can store different data types
- Written inside parentheses ()

Example:

```
t = (1, "Hello", 3.5, True)
```



Creating a Tuple

A tuple can be created by placing values inside parentheses.

Example:

```
tuple1 = (1, 2, 3)
tuple2 = ("apple", "banana", "mango")
tuple3 = (10, "Python", 5.5)
```

Single element tuple:

```
single = (5,)
```



Accessing Tuple Elements

Tuple elements are accessed using index numbers.

Example:

```
t = ("apple", "banana", "mango")
```

```
print(t[0])
```

```
print(t[1])
```

Output:

```
apple
```

```
banana
```

```
hello_world.py > ...
1  t = ("apple", "banana", "mango")
2  print(t[0])
3  print(t[1])
4
5
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL
PS C:\Users\prati\Downloads> & C:/Users/
apple
banana
PS C:\Users\prati\Downloads>
```



Tuple Operations

- Some operations that can be performed on tuples:
- Concatenation (+)
- Repetition (*)
- Membership (in / not in)

Example:

```
t1 = (1,2)
```

```
t2 = (3,4)
```

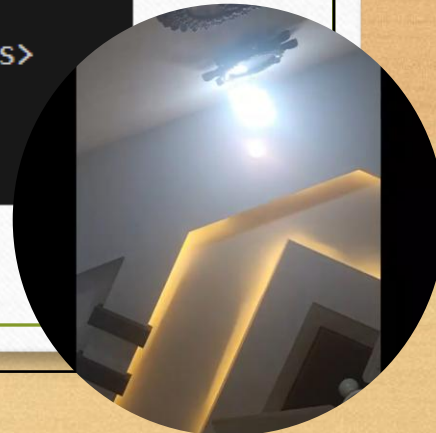
```
print(t1 + t2)
```

Output:

```
(1,2,3,4)
```

```
hello_world.py X
hello_world.py > ...
1  t1 = (1,2)
2  t2 = (3,4)
3
4  print(t1 + t2)
5
6
```

```
PS C:\Users\prati\Downloads> & C:/Python/python hello_world.py
(1, 2, 3, 4)
PS C:\Users\prati\Downloads>
```



Tuple Methods

Python has two built-in tuple methods:

`count()` – counts how many times a value appears

`index()` – finds the index of an element

Example:

```
t = (1,2,3,2,2)
```

```
print(t.count(2))
```

```
print(t.index(3))
```

Output:

```
hello_world.py > ...  
1  t = (1,2,3,2,2)  
2  
3  print(t.count(2))  
4  print(t.index(3))  
5  
6
```

```
PS C:\Users\prati\Downloads> & C:  
3  
2  
PS C:\Users\prati\Downloads>
```



Advantages of Tuple

- Faster than lists
- Protects data (cannot be modified)
- Used for fixed data
- Requires less memory

Conclusion

- Tuples are an important Python data structure.
- They store multiple values in a single variable.
- Tuples are ordered, immutable, and efficient.
- Used when data should not be changed.

